

Clinical Evaluation of Arthroscopically Assisted Anterior Cruciate Ligament Reconstruction: Patellar Tendon Versus Gracilis and Semitendinosus Autograft

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Purpose: To evaluate the clinical results of anterior cruciate ligament (ACL) reconstruction in patients with ACL-deficient knee in 2 similar groups of patients. **Type of Study:** A prospective randomized comparison of patellar tendon (PT) versus semitendinosus and gracilis tendon (STG) autografts for ACL reconstruction. **Methods:** Between 1994 and 1996, 85 consecutive male patients with chronic ACL-deficient knees underwent arthroscopically assisted reconstruction with either autologous PT or double-loop STG (4-strand) graft. PT grafts were used in patients with even-numbered birth dates and STG grafts for those with odd-numbered birth dates. Preoperatively, no significant differences between the 2 groups were noted with respect to age, level of activity, and degree of laxity (χ -square analysis). A standardized rehabilitation program was used for both groups postoperatively that included immediate active extension and early weight bearing and gradual flexion. Return to sports was permitted 8 months postoperatively. Assessment of the patients was carried out using a questionnaire, clinical assessment, Lysholm knee scores, the International Knee Documentation Committee scale, and radiological examination. **Results:** At a mean follow-up of 81 months, there was no significant difference between the 2 groups with respect to subjective complaints (recurrent giving way, functional level) or objective laxity evaluation, including KT-1000 measurement or return to sports. Loss of extension of $\leq 5^\circ$ was greater in the PT group (12 patients, 30%) than in the STG group (8 patients, 17%). There was loss of flexion of $\leq 15^\circ$ in 5 patients (12%) in the PT group and 1 patient (2.2%) in the STG group. Anterior knee pain was recorded in 10 patients (24%) in the PT group and 3 patients (5%) in the STG group. The Lysholm knee score was 91.6 and 92.7 for the PT and STG groups, respectively, and the Tegner activity score decreased from 8.9 preoperatively for both groups to 7.9 for the PT group and 7.8 for the STG group. **Conclusion:** In this study, the 2 groups had comparable results in terms of patient satisfaction, activity level, and knee function. Our study showed that patellofemoral problems and loss of knee motion are more frequent in patients with PT grafts than in those with STG grafts. **Level of Evidence:** Level I, Randomized Controlled Trial. **Key Words:** Anterior cruciate ligament—Patellar tendon—Gracilis and semitendinosus—Muslim—EndoButton.

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Various intra-articular substitutions for the anterior cruciate ligament (ACL) have been attempted. Intra-articular reconstructions with autograft, allogenic graft,¹ and artificial ligaments^{2,3} have been used. Autologous grafts are used widely for chronic instability and satisfactory results have been reported. The most frequently used grafts are patellar tendon (PT)⁴⁻¹⁰ and semitendinosus and gracilis (STG).¹¹⁻¹⁶ The advantages and disadvantages of both grafts are well known. Although prosthetic devices and allo-

grafts are attractive as ACL substitutes because they eliminate the loss of the patient's own tissue, potential complications do exist.^{3,4} At present, autologous grafts remain the most popular form of intra-articular replacement of the ACL. Isometric positioning of the graft, use of strong tissue, adequate graft fixation, staged rehabilitation with early passive extension, and arthroscopic technique can be expected to improve the results of ACL reconstruction and reduce associated complications.¹⁷⁻²¹

The purpose of this study was to evaluate the clinical results of ACL reconstruction using 2 different grafts commonly used to reconstruct the ACL in patients with ACL-deficient knee in 2 similar groups of patients. The same surgical technique, rehabilitation program, and methods of follow-up were used for both groups. All procedures were performed or supervised by the senior author (S.A-R.I.).

METHODS

From 1994 to 1996, 110 consecutive patients with an ACL-deficient knee underwent arthroscopically assisted ACL reconstruction with either autologous PT or double-loop (4-strand) STG tendon. Of the 85 patients available for evaluation, 40 patients had their ACL reconstructed with autologous PT graft and 45 patients had their ACL reconstructed using STG graft. All the patients were male with an average age of 22.3 years (range, 17 to 34 years) at the time of surgery. The majority of injuries occurred while playing football (75 patients), basketball (6 patients), and handball (4 patients). The mechanism of injury was external rotation and valgus injury (47 patients) and hyperextension injury (15 patients); in 23 patients, the mechanism of injury was not determined. The main symptoms of the patients before surgery were giving way during daily activity (75 patients) and during sports (8 patients), and recurrent locking (2 patients). All patients were recreational or competitive athletes; no professional athletes were treated in our series (there are no professional athletes in Kuwait). There were 30 recreational and 10 competitive athletes in the PT group and 32 recreational and 13 competitive athletes in the STG group. Competitive athletes were defined as active members of a club team who participated regularly in competition. Using these criteria for competitive and recreational athletes, no statistical differences were noted with respect to the age (χ -square analysis). Clinical examination showed that results of the Lachman test, anterior drawer sign, and pivot-shift test were positive in all patients. Ten patients had

valgus laxity at 30° flexion (6 PT and 4 STG). No patients had associated posterior tibial translation or posterolateral instability. No statistical differences in preoperative laxity were observed (*t* test). Preoperative radiographs revealed 13 patients (5 PT and 8 STG) with narrowing of the medial compartment and patellofemoral changes. The indications for surgery were functional instability during daily activity, instability during sports, and symptoms of locking and recurrent effusion caused by an associated torn meniscus. At surgery, 43 patients had a midsubstance tear of the ACL, 15 at the femoral attachment, and 5 near the tibial attachment. In 22 patients, the site of the torn ACL was difficult to determine. Tears of the medial meniscus were found in 20 patients and the lateral meniscus in 10 patients. Chondral lesions of the medial femoral condyle were noted in 6 patients. The articular surfaces of the patella was affected in 18 patients in the form of softening and fissuring.

OPERATIVE TECHNIQUE

Harvesting the Graft

After the knee was examined under anesthesia, a pneumatic tourniquet was inflated. When the STG graft was used, the 2 tendons were separated from their insertion and harvested using a tendon stripper. In the case of a PT graft, the middle third of the PT was sharply incised by parallel incision, creating a 10-mm wide graft. The 2 blocks of bone each measured 2.5 to 3 cm in length.

Transplant Preparation and Endoscopic Technique for ACL Reconstruction

Arthroscopy was performed through the anterolateral and anteromedial portals while the second team was preparing the graft. Resection of any meniscal tear and ACL remnants was performed, as was routine notchplasty. The tibial tunnel was made using a guidewire placed at the isometric point of the tibial attachment of the ACL. The isometric tunnel site on the femur was determined by using a femoral guide. The guidewire was inserted through the tibial channel into the selected area on the femur through the tibial channel over the guide; the femur was drilled 6 mm deeper than the desired insertion of the graft to provide the turning radius for the EndoButton (Smith & Nephew, Andover, MA). The 4.5-mm EndoButton drill bit was used to overdrill the passing pin, specifically noting the total length at the moment of cortical breakout. A

calibrated depth probe was used to accurately measure the femoral channel length.

Fixing the Graft

The STG graft polyester tap was threaded through the loop. The sutures were fed through the middle 2 holes of the femoral suture plate. The outer 2 holes were threaded with 2 further strong sutures (a pull and a flip suture). The knot length was equal to the femoral canal length minus the transplant length, which come to lie in the femoral channel. The graft was inserted and pulled through the knee under arthroscopic guidance to the mark on the tendon. The flip suture was pulled—this turns the suture plate parallel to the femur. The tibial sutures pulled the graft back 7 mm so the suture plate came down to lie on the lateral cortex of the femur. For the PT graft, the same principles were applied, taking into consideration that the bone blocks should be situated within the drill channel. The distal end of the graft was fixed with screw and spiked washer or with a small plate, 2 screws, and staple, with the knee in full extension. For the PT graft, the bone blocks were fixed with interference screws.

Postoperative Care

Postoperatively, the knee was immobilized in a well-padded dressing in full extension and ice packs were used. One gram of ceftriaxone sodium was administered intravenously preoperatively and on the second postoperative day. Active assisted motion exercises were started on the second postoperative day. Initially, range of motion was from full extension to 30° of flexion. Partial to full weight bearing was allowed on the second or the third postoperative day.

RESULTS

We evaluated 85 patients at a mean follow-up of 81 months (range, 60 to 96 months). The time from injury to surgery was 4 to 13 months (average, 9.7 months). The hospital stay averaged 5.9 days (range, 5 to 9 days).

Subjective Results

The subjective results of the ACL reconstructions were evaluated using a detailed questionnaire completed by each patient. All patients were able to return to their previous employment or to school at an average of 4.6 months (range, 2 to 10 months). In the PT group, 35 patients (78.5%) reported overall good results, indicating that they were satisfied with surgery

versus 41 patients (91%) in the STG group. Overall, 78 patients did not experience any episodes of giving way: 37 patients (92.5%) in the PT group and 41 patients (91%) in the STG group. Occasional giving way was reported by 2 patients (5.5%) in the PT group and 3 patients (6.6%) in the STG group. Two patients reported giving way on activity (1 PT and 1 STG). Anterior knee pain was reported by 13 patients (15.5%): 10 PT patients (25%) and 3 STG patients (6.6%). In the PT group patients, pain was anterior, especially while kneeling to pray and when sitting cross-legged on the floor. Sixty-six patients (77.6%) returned to their preinjury level of activity; 10 patients (6 PT and 4 STG) were able to participate at a lower level of activity, and 9 patients chose not to participate in sports because of symptoms of giving way, pain, or apprehension about possible reinjury. There was no statistical difference between the 2 groups with regard to return to activity.

Objective Results

Physical examination showed that 2 PT graft patients had a loss of 10° of extension. Twenty patients (33.5%) had a loss of ≤5° of extension (12 PT, 30%; and 8 STG, 17.7%). Six patients had loss of flexion (5 PT and 1 STG) up to 15° as compared with the uninvolved knee. The Lachman test result was negative in 74 patients—35 PT (87.5%) and 39 STG (86.6%)—and positive in 12 patients (5 PT and 7 STG). The anterior drawer sign was positive in 11 patients (5 PT and 7 STG). Thirteen patients (5 PT and 8 STG) developed a pivot glide; 13 patients had a positive friction test result with retropatellar tenderness (the same patients who complained of anterior knee pain). Instrumental laxity testing using the KT-1000 arthrometer revealed that 35 PT patients (87.5%) and 38 STG patients (84.4%) had an injured-to-uninjured knee difference of ≤3 mm (Table 1). There was no statistically significant difference between the 2 grafts with respect to the average values compiled by arthrometer testing. The Lysholm knee score²² (Table 2) was better with the STG graft than the PT graft; it was 92.7 points (range, 81–97) for STG and 91.6 points (range, 79–96) for PT grafts. The average Tegner activity score (Table 3) decreased from 89 before surgery to 7.9 for PT and 7.8 for STG at follow-up. The International Knee Documentation Committee scale for the PT graft showed 25 patients (62.5%) had grade A excellent results, 10 patients (25%) had grade B good results, and 5 patients (12.5%) had grade C fair results. No patient had a grade D poor result. Of

TABLE 1. Ligament Testing Results

Test	No. of Patients	
	PT	STG
Lachman		
Negative	35 (87.5%)	39 (86.6%)
Grade 1	5 (10)	7 (17.7%)
Anterior drawer		
Negative	35 (87.5%)	38 (84.4%)
Grade 1	5 (12.5%)	7 (17.7%)
Pivot-shift		
Negative	35 (87.5%)	37 (84.2%)
Pivot-glide	5 (12.5%)	8 (17%)
KT-1000 maximum manual (1 N)		
0-2 mm	35 (87.5%)	38 (84.4%)
3-5 mm	3 (7.5%)	3 (6.6%)
>5 mm	2 (5%)	4 (8.8%)

the patients with STG grafts, 28 patients (62.5%) had grade A excellent, 10 patients (22%) had grade B good, and 7 patients (15.5%) had grade C fair results. No patient had grade D poor.

Radiographic examination showed moderate disease of the patellofemoral cartilage in 8 PT graft patients (20%) versus 3 STG graft patients (6.6%). The medial compartment showed moderate degenerative changes in 6 patients (15%) in the PT group and 8 patients (17.7%) in the STG group. Patients who had degenerative changes in the medial compartment had a chondral lesion or partial meniscectomy or both.

DISCUSSION

In 1994, we began arthroscopically assisted ACL reconstruction using PT or STG (4-strand) as graft material. In this study, the results of the ACL reconstruction with either PT or STG were comparable to each other. Noyes et al.²³ reported both semitendinosus and gracilis had the strength of 70% and 49%, respectively, of the intact ACL. Theoretically, STG tensile strength will be doubling them (4 strands). Both a doubled STG graft and PT graft should have

TABLE 2. Lysholm Knee Score

Score (points)	No. of Patients	
	PT	STG
Excellent (95-100)	20 (50%)	24 (53%)
Good (84-94)	16 (40%)	18 (40%)
Fair (65-83)	4 (10%)	3 (7%)

TABLE 3. Tegner Activity Score

PT			STG		
Preop	Postop	No. of Patients	Preop	Postop	No. of Patients
8-9	7-9	23 (57.5%)	8-9	7-9	26 (57.7%)
7-8	6-8	15 (37.5%)	7-8	6-8	16 (35.5%)
6-7	5-6	2 (5%)	6-7	5-6	3 (6.8%)

tensile strength comparable to the normal ACL. Interference fit fixation of the bone of the PT graft has been known to be superior to other types of fixation.^{8,24} We used a post and spike washer and staple for distal fixation in the STG graft group and interference screws in the PT group. We did not experience any failures of fixation. Our methods of fixation were satisfactory for immediate active extension and flexion of the knee and early weight bearing. Recently, early weight bearing after ACL reconstruction with the PT graft and interference screw fixation^{25,26} has been discussed. Chiroff²⁷ reported in his histologic study that weight bearing may not be advisable after reconstruction with an STG graft because periosteal fibrous continuity is not evident until 4 weeks post-operatively, so that fixation by means other than interference screws may not be able to withstand the loads imposed on the ACL on weight bearing.^{8,24} Immediate active flexion and extension and weight bearing was advised by others regardless of the type of graft.^{15,16,28,29}

We did not find any significant difference between the 2 groups with respect to residual anterior tibial translation as determined by anterior drawer test, Lachman test, pivot-shift test, or with the KT-1000 arthrometer. Although not statistically significant, 87.5% of patients with PT grafts had a KT-1000 difference of ≤ 2 mm compared with 84.4% of patients with STG grafts. Marder et al.²⁹ reported no significant difference between the 2 groups with respect to Lachman or pivot-shift testing or with the arthrometric results. They reported that 86% of patients with a PT graft had a KT-1000 difference of ≤ 2 mm compared with 74% of patients with the STG graft. Aglietti et al.³⁰ reported that the difference in anterior displacement and pivot-shift between the 2 groups was not statistically significant, but that there was a trend toward better stability with the PT graft. They stated that weakness of the STG graft may be in the initial fixation where a staple could not be used in 6 cases because of insufficient length, and they ad-

vised a less accelerated rehabilitation program. Harter et al.³¹ found that the PT and STG grafts were equally efficient in limiting anterior tibial translation as measured by the KT-1000.

Holmes et al.³² reported the incidence rate of patellofemoral pain after arthroscopically assisted ACL reconstruction with autologous PT graft is approximately 16% to 47% versus 3% to 21% for the STG graft.^{29,33-36} Marder et al.²⁹ reported that 17 of 72 patients (24%) (11 PT and 6 STG) complained of anterior knee pain with activity. Aglietti et al.³⁰ reported that patellofemoral pain and crepitus are more frequent after use of a PT graft. Our finding was that patellofemoral pain and crepitus were more common in the PT group (24%) than in the STG group (5%). This was reported by the patients on activity and specially on kneeling (during praying) and was confirmed on clinical examination. This is relatively in agreement with Marder et al., but there were significantly more patients with the PT graft who had patellofemoral pain in our group. Shaieb et al.³⁷ reported no significant difference between the 2 groups with regard to Lysholm knee score, reduction of activity, KT-1000 arthrometer reading, or return to sports, but significantly more patients with the PT graft had patellofemoral pain (42% v 20% at 3 years follow up). Corry et al.³⁸ reported that pain on kneeling was higher in the PT group. Pinczewski et al.³⁹ reported that arthroscopic ACL reconstruction with either graft results in a similar outcome, restoring stability over a 5-year period. They also reported that patients with PT grafts are at a greater risk of developing early signs of osteoarthritis and fixed flexion deformity (18%), but there was no significant increase in the STG graft group between 2 and 5 years and there was no significant difference between 2 groups.

After 3 years of follow-up, Beynnon et al.⁴⁰ reported that the objective results of ACL replacement with PT were superior to those replaced with 2-strand STG grafts with regard to knee laxity, pivot-shift, and strength of knee flexor muscles. They also reported that patients with 2 strands of STG had greater anterior laxity than PT grafts, as 41% of patients with STG grafts had a positive pivot-shift compared with 14% of patients with PT grafts. However, the 2 groups had comparable results in terms of patient satisfaction, activity level, and knee function. Beynnon et al. reported that the ACL reconstruction used 2 strands of STG graft versus PT graft, whereas in our study and others, 4 strands of STG versus PT graft were used.

Loss of flexion and extension in our series was more frequent in the PT group than in the STG group. In

spite of proper, isometric placement of the tunnels, we noticed in those patients, except in 4 cases (3 PT and 1 STG), that their tunnels were anterior; in the STG group maybe their grafts had been stretched during rehabilitation. Aglietti et al.³⁰ reported that almost half of the PT graft group had minimal extension loss, which appears to be of little significance. Marder et al.²⁹ reported 19 of 72 patients had loss of motion, which was more frequent in the PT group than in the STG group. They reported that 5 patients had a flexion contracture of greater than 5°. In our series, we found that loss of motion, and in particular loss of flexion and anterior knee pain, were the common complaints of our patients.

CONCLUSIONS

Our study confirms the clinical impression that ACL reconstruction with PT or STG graft restores knee stability and allows return to a high level of functional and sports activity. There was no overwhelming difference between the 2 grafts with regard to objective stability, but there was an increase in the incidence of early degenerative changes both clinically and radiologically; fixed flexion and extension deformity and patellofemoral pain in patients with a PT graft are at a greater risk of developing osteoarthritis over a period of more than 3 years. We believe that the STG graft is preferable to the PT graft, especially for Muslim patients for whom kneeling to pray is a daily requirement and because people in this part of the world usually sit cross-legged on the floor.

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